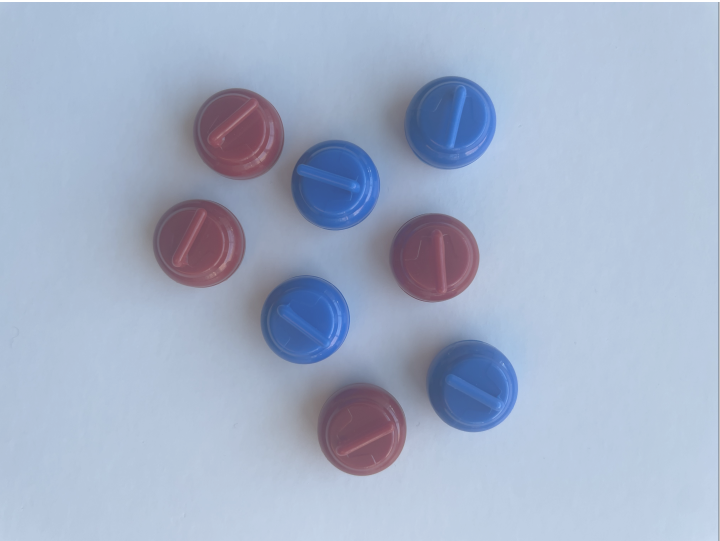


Project: Segment an Image

[My Solutions >](#)

For this question, you'll be working with the following image of a set of curling stones. The image is included with course files so you can work in MATLAB. Your goal is to create a mask which segments the curling stones from the background in order to measure features of the stones.



For this problem, you are given two lines of code which read in the image and convert it to grayscale. You will need to add your own code to provide a mask which accurately segments the curling stones from the background. Use variable name `curlingMask`.

NOTE: If you use an app to segment the image in MATLAB, generate your segmentation function and *copy all lines into the bottom of your solution* code here to allow Grader to run it. For example:

```
1  curlingImg = imread("curlingImage.jpg");
2  grayImg = im2gray(curlingImg);
3
4  % Add your code below
5
6  % Call segmentation function
7  curlingMask = yourSegmentationFunction(curlingImg);
8
9  % Generated segmentation function
10 function [BW,maskedImage] = yourSegmentationFunction(X)
11     % Function code
12 end
```

Script ?

Save

Reset

MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1  curlingImg = imread("curlingImage.jpg");
2
3  % Add your code below
4  grayImg = im2gray(curlingImg);
5  curlingMask = segmentImage(grayImg);
6
7  function [BW,maskedImage] = segmentImage(X)
8
9  % Threshold image - adaptive threshold
10 BW = imbinarize(im2gray(X), 'adaptive', 'Sensitivity', 0.390000, 'ForegroundPolarity', 'dark');
11
12 % Invert mask
13 BW = imcomplement(BW);
14
15 % Fill holes
16 BW = imfill(BW, 'holes');
17
18 % Fill holes
19 BW = imfill(BW, 'holes');
20
21 % Close mask with default
22 radius = 25;
23 decomposition = 0;
24 se = strel('disk', radius, decomposition);
25 BW = imclose(BW, se);
26
27 % Create masked image.
28 maskedImage = X;
29 maskedImage(~BW) = 0;
30 end
31
32
```

Run Script ?

Assessment: All Tests Passed

✔ Is the binary mask a valid mask for this image?

✔ Does the mask successfully separate the curling stones from the background?

Output

Code ran without output.